Training Day 3

1. Topics Covered and Brief Explanation

1. Supervised Learning

Supervised Learning is a type of machine learning in which a model is trained using labelled data. Each training example contains input data along with its correct output. The model learns the relationship between them and uses it to predict outputs for new data.

Examples: Email spam detection, house price prediction and student result prediction.

2. Unsupervised Learning

Unsupervised Learning is a type of machine learning in which a model works with unlabelled data. It identifies hidden patterns, similarities and groups within the dataset without predefined output labels.

Examples: Customer segmentation, grouping similar products and analysing shopping patterns.

3. Reinforcement Learning

Reinforcement Learning is a type of machine learning in which an agent learns by interacting with an environment. It receives rewards for correct actions and penalties for incorrect actions, helping it improve its decisions over time.

Examples: Robotics, game-playing systems and autonomous navigation.

4. Difference Between the Three Types

Type	Training Data	Main Purpose
Supervised Learning	Labelled data	Prediction
Unsupervised Learning	Unlabelled data	Pattern identification
Reinforcement Learning	Feedback through rewards and penalties	Learning through interaction

2. Tools and Technologies Used

Tool	Purpose
Python	Basic programming and understanding ML examples.
Google Colab	Writing and executing Python code online.
NumPy	Performing basic numerical operations.
Google Chrome	Exploring machine learning concepts and examples.

3. Practical Learning / Code Example

Explored a simple example of Supervised Learning using Python to understand how a model learns from labelled data.

Example: Predicting student marks using Linear Regression

```
from sklearn.linear_model import LinearRegression


# Training data
X = [[1], [2], [3], [4], [5]]
y = [20, 40, 60, 80, 100]
# Creating the model
model = LinearRegression(
# Training the model
model.fit(X, y)
# Making a prediction
prediction = model.predict([[6]])
print("Predicted Marks:", prediction[0])
```

Explanation:

- LinearRegression() creates a regression model.
- fit() trains the model using the given data.
- predict() estimates the output for new input data.
- The model predicts approximately 120 marks for 6 hours of study based on the sample relationship.

Note: This is a simplified demonstration using sample data, not a realistic student performance model.

4. Learning Outcomes

- Understood the three major types of Machine Learning.
- Learned the difference between labelled and unlabelled datasets.
- Understood how reinforcement learning uses rewards and penalties.
- Gained basic exposure to implementing a supervised learning model using Python.
- Identified practical applications of different ML techniques.